

Time Series Forecasting

“Forecasting is not about predicting the future, it’s about reducing uncertainty.” -- Paul Saffo

Agenda

- Introduction to Forecasting Methods
- Simple Moving Average (SMA)
- Weighted Moving Average (WMA) & Centered Moving Average (CMA)
- Building Programs for Moving Average Models in Python
- Simple Exponential Smoothing (SES)
- Double Exponential Smoothing (Holt's Model)
- Triple Exponential Smoothing (Holt-Winters Model)
- Building Programs with Exponential Smoothing (SES & Holt)
- Building Programs with Holt-Winters & Model Evaluation
- Model Evaluation & Selection
- Advanced Topics, Practical Considerations & Q&A

Introduction to Forecasting Methods

Two Broad Categories

1. Qualitative Forecasting

- Based on judgment, opinions, and experience.
- Used when historical data is unavailable or unreliable.
- Common methods:
 - Delphi Method
 - Market Research
 - Jury of Executive Opinion
 - Sales Force Composite

2. Quantitative Forecasting

- Based on numerical data and statistical models.
- Used when consistent, relevant historical data is available.

Types:

- Time Series Methods
- Machine Learning Methods

Quantitative Forecasting

Time Series Models

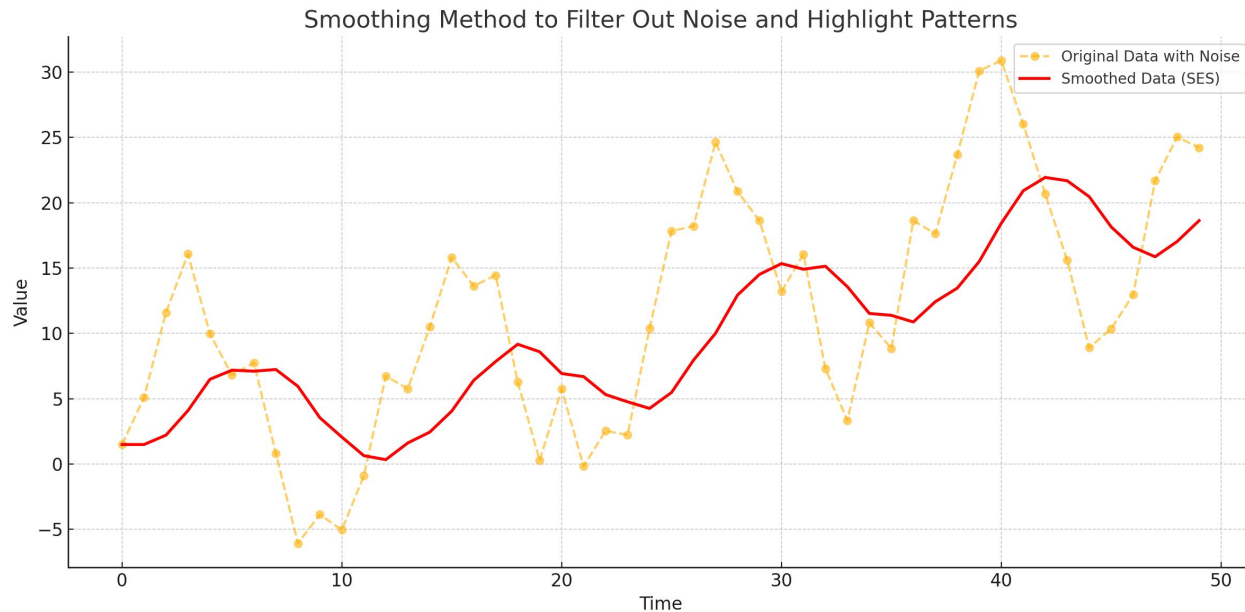
- Analyzes patterns in a single variable over time.
- Assumes past patterns (trend, seasonality, cycles) continue.
- **Key Methods:**
 - Smoothing Techniques
 - Decomposition
 - ARIMA Models

Machine Learning:

- Tree-Based Models: Random Forest, XGBoost
- SVM & SVR: Captures complex patterns
- Neural Networks (LSTM, RNNs, Transformers): Handles sequential dependencies

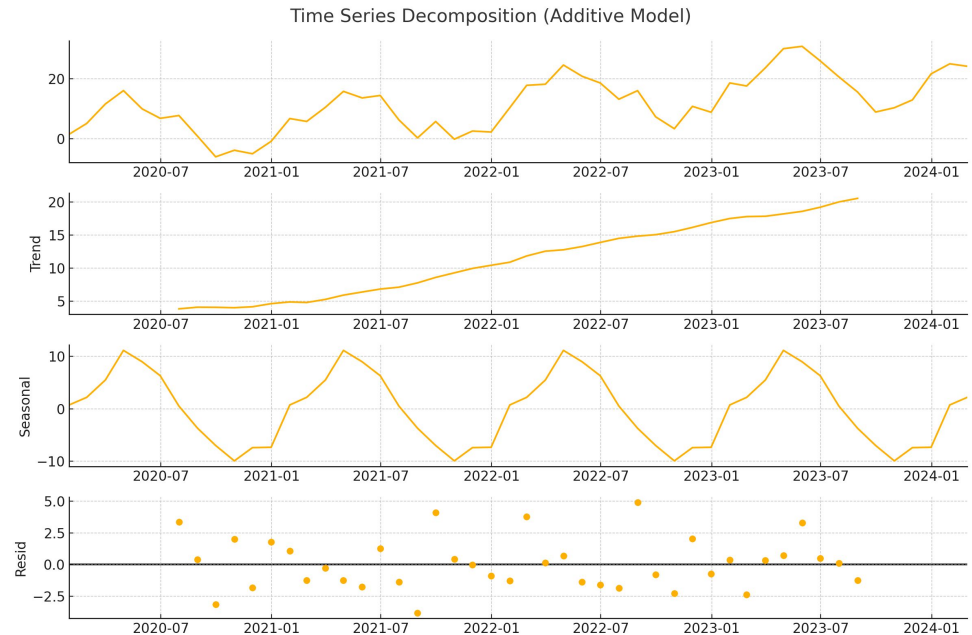
Smoothing Methods

- Purpose:
 - Filter out noise to highlight underlying patterns.
- Types:
 - Simple Moving Average
 - Weighted Moving Average
 - Centered Moving Average
 - Moving Average Exponential Smoothing (Simple, Holt, Holt-Winters)



Decomposition & ARIMA

- Decomposition:
 - Breaks down time series into:
 - Trend
 - Seasonal
 - Cyclical
 - Irregular components



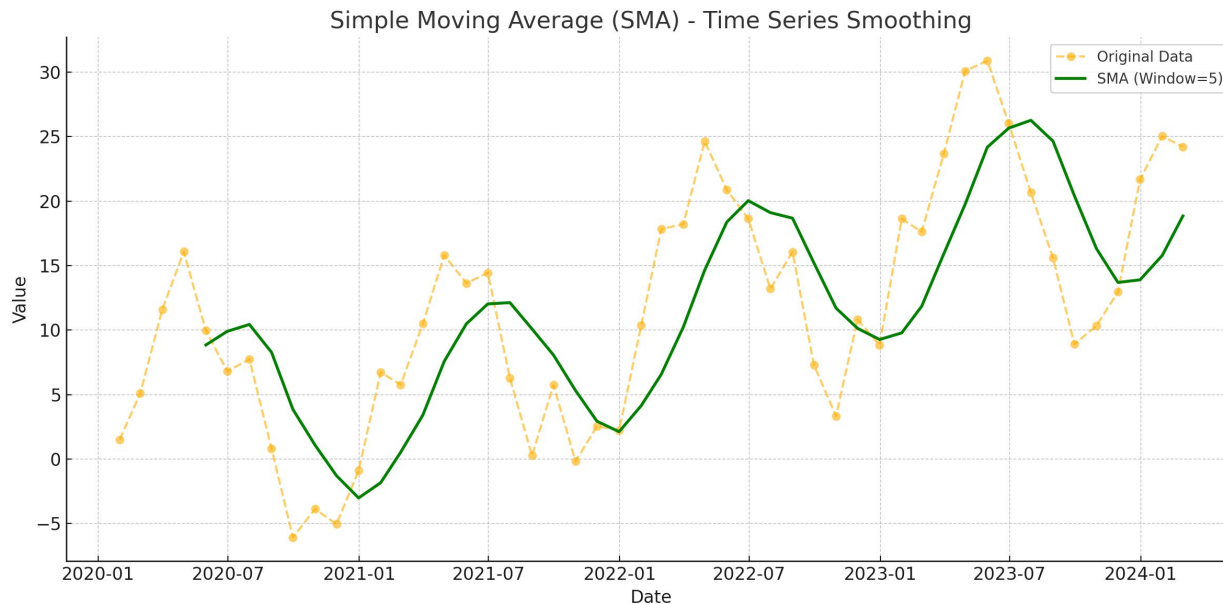
- ARIMA Models:
 - Combines Auto Regression (AR), Differencing (I), and Moving Average (MA).
 - Extensions: SARIMA, ARIMAX

Simple Moving Average

- The Simple Moving Average (SMA) is one of the most fundamental techniques in time series forecasting.
- Its core idea is to smooth out short-term fluctuations and highlight longer-term trends or cycles by averaging data over a fixed number of previous time periods.

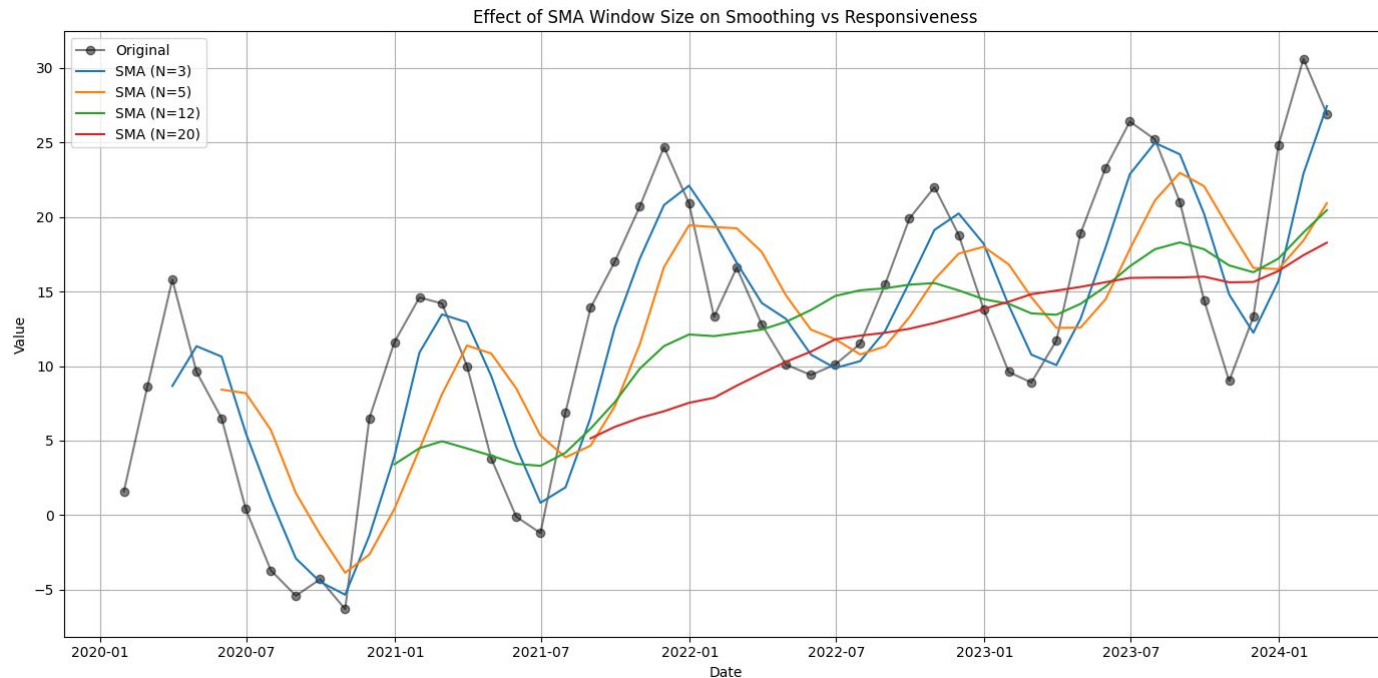
$$F_t = \frac{Y_{t-1} + Y_{t-2} + \cdots + Y_{t-N}}{N}$$

$$F_t = \frac{1}{N} \sum_{i=1}^N Y_{t-i}$$



Simple Moving Average

- The parameter N defines the window size — how many past observations are considered for the average.
- A small N (e.g., 3 or 5) gives more responsive forecasts, but may retain noise.
- A large N (e.g., 12 or 20) gives smoother forecasts, but may lag behind sudden changes
- Choosing N is a trade-off between smoothness and responsiveness.



SMA Strengths:

Simplicity: Easy to understand and implement, even with basic programming or spreadsheets.

Noise Reduction: Effective in smoothing random fluctuations in data.

No Need for Complex Assumptions: Doesn't require the data to follow a specific statistical distribution.

SMA Weakness

Key Weaknesses

- **Lagging Trends:**
 - The SMA is a "lagging indicator." When a time series has a clear upward or downward trend, the SMA forecast will consistently be behind the actual data.
- The forecast will always be lower than the actual data in an upward trend and higher in a downward trend.
- **Equal Weighting of Observations:**
 - The model gives equal importance to all data points within the window N
- Inability to Handle Seasonality (Refer Notebook)
- Parameter Selection

Weighted Moving Observations

- The Weighted Moving Average enhances SMA by assigning different weights to each of the past N observations
- Usually gives more weight to recent observations and less weight to older ones.

Let:

- Y_{t-i} : Actual value at time $t - i$
- w_i : Weight assigned to Y_{t-i} , such that $\sum_{i=1}^N w_i = 1$
- F_t : Forecasted value at time t

$$F_t = \sum_{i=1}^N w_i \cdot Y_{t-i}$$

Where:

- w_1 is the weight for the most recent observation (e.g., yesterday),
- w_N is the weight for the oldest observation in the window,
- And $w_1 > w_2 > \dots > w_N$ to emphasize **recency**.

Strengths of WMA

- **More Sensitivity to Trends:** Since recent data is weighted more, WMA can better capture recent upward or downward trends than SMA.
- **Flexible Design:** Allows custom emphasis based on known data relevance.
- **Smoother than Raw Data:** Still acts as a smoother, but with more realistic responsiveness than SMA.

Weaknesses of WMA

- **Complexity in Weight Selection:** Unlike SMA, weights are not automatic — choosing them requires insight, tuning, or experimentation.
- **Still Linear:** WMA doesn't model non-linear relationships or seasonality explicitly.
- **Limited Forecast Horizon:** Like SMA, WMA is better for short-term forecasting and does not perform well for long-term projections.

Centred Moving Average

- The Centered Moving Average (CMA) is a smoothing technique used in time series analysis for trend estimation but not for forecasting
- For a 3-point CMA, the average for time t is calculated using the values at time $t-1$, t , and $t+1$
- By Choosing Window Size should be equal to the length of seasonal cycle. CMA averages out seasonal variations and random noise

Example for N

- $N=12$ for monthly data with annual seasonality
- $N=4$ for quarterly data

$$CMA_t = \frac{Y_{t-1} + Y_t + Y_{t+1}}{3}$$

Limitations of SMA and WMA

Lag in Response to Trends

- SMA and WMA are inherently backward-looking: they use past observations to compute the forecast for the next period.
- As a result, they lag behind actual trends. For example, if the data is trending upward, both SMA and WMA will underestimate future values because they average in older, lower values.
- The lag becomes especially problematic when the trend changes direction abruptly.

Equal or Fixed Weighting Issues

- SMA assigns equal weight to all values in the window. This implies that a value from 5 weeks ago is just as important as one from last week, which may not reflect reality.
- WMA allows for differentiated weighting, but the weights must be manually specified and remain fixed for each window. This creates two issues:
 - Subjectivity: Choosing optimal weights is arbitrary and often lacks a systematic basis.
 - Rigidity: Fixed weights do not adjust based on changing patterns in the data.

Limitations of SMA and WMA

- Poor Long-Term Forecasting
 - Neither SMA nor WMA are designed for multi-step-ahead forecasting.
 - They don't model trends or seasonality explicitly, which are often essential for medium-to-long-range planning
- Sensitivity to Window Size (N)

The choice of N significantly affects the forecast but lacks a robust selection criterion.

Small N leads to high variability (less smoothing), while large N can over smooth and miss important shifts.

Exponential Smoothing technique

Simple Exponential Smoothing

$$F_t = \alpha Y_{t-1} + (1 - \alpha)F_{t-1}$$

Where:

- F_t is the forecast for time t
- Y_{t-1} is the actual observed value at time $t - 1$
- F_{t-1} is the forecast made for time $t - 1$
- α is the **smoothing parameter**, $0 < \alpha < 1$

Example:

- $\alpha = 0.8 \rightarrow$ The model gives 80% weight to the most recent actual value and only 20% to the previous forecast.
- $\alpha = 0.2 \rightarrow$ The model gives only 20% weight to the latest observation, favoring stability over responsiveness.

Value of α	Interpretation	Behaviour
Close to 1	Very responsive	Forecast quickly adjusts to recent changes (but may be noisy)
Close to 0	Very smooth	Forecast changes slowly (lags behind current data)

SES Limitations

- SES assumes the data is flat or stable over time — i.e., no trend.
- If your data exhibits a consistent upward or downward movement, SES will fail to capture the evolving pattern.
- To solve the above limitation, we will go for Holts method

Double exponential Smoothing Holts method

Holts method have two components:

- **Level Component:** This is similar to SES and captures the most recent average value of the data.
- **Trend Component:** This is a new, separate component that estimates the slope or rate of change in the data.

Holts method: Level component

The Level Component (L_t):

The smoothed estimate of the current value of the series.

$$L_t = \alpha Y_t + (1 - \alpha)(L_{t-1} + b_{t-1})$$

- Y_t is the actual data point at time t .
- $(L_{t-1} + b_{t-1})$ is the one-step-ahead forecast from the previous period, which is the previous level plus the previous trend.
- α is the smoothing parameter for the level

Holts method: Trend Component

The Trend Component (b_t):

The smoothed estimate of the rate of change (upward/downward) at time t .

$$b_t = \beta(L_t - L_{t-1}) + (1 - \beta)b_{t-1}$$

- $(L_t - L_{t-1})$ is the most recent observed change in the level of the series.
- b_{t-1} is the trend estimate from the previous period.
- β is the smoothing parameter for the trend.

Holts Method: forecast

$$F_{t+m} = L_t + m \cdot b_t$$

F_{t+m} : Forecast made at time t for m steps into the future

L_t : the estimated level at time t ,

b_t : the estimated trend at time t ,

m : the number of steps ahead.

Role of alpha and Beta

- **Alpha (α):** Controls the smoothing of the level component.

- A **high α** (closer to 1) gives more weight to the most recent data point. The level will react quickly to new values, making it more responsive to short-term changes and potentially more volatile.
- A **low α** (closer to 0) gives more weight to past data. The level will be much smoother and less reactive to new, short-term fluctuations.

- **Beta (β):** Controls the smoothing of the trend component.

- A **high β** (closer to 1) gives more weight to the most recent change in the level ($L_t - L_{t-1}$). The trend will adjust quickly to new directional shifts in the data. This is useful for volatile or quickly changing trends but can make the forecast's slope erratic.
- A **low β** (closer to 0) gives more weight to the historical average trend. The trend component will be stable and slow to react to new changes, which is ideal for a consistent, long-term trend.

When to use Holt's method

- Time series with a linear trend but no seasonality.
- Forecasting sales, stock prices, temperature, energy consumption, etc., where upward or downward momentum exists

Limitations

- Holts method does not account for seasonality—the repeating patterns or fluctuations that occur at regular intervals (e.g., hourly, daily, monthly).

Holt Winters method

- The Holt-Winters Model, also known as Triple Exponential Smoothing
- Holts Winter = Holts+ Seasonal component

It is specifically designed to model:

- **Level** (baseline value of the series)
- **Trend** (upward or downward direction over time)
- **Seasonality** (repeating patterns over time)

Holt-Winters Formulas

Component	Equation
Level	$L_t = \alpha(Y_t - S_{t-m}) + (1 - \alpha)(L_{t-1} + b_{t-1})$
Trend	$b_t = \beta(L_t - L_{t-1}) + (1 - \beta)b_{t-1}$
Seasonal	$S_t = \gamma(Y_t - L_{t-1} - b_{t-1}) + (1 - \gamma)S_{t-m}$
Forecast	$F_{t+h} = L_t + h \cdot b_t + S_{t+h-m(k)}$

k = floor((h - 1)/m): ensures proper seasonal alignment.

m denotes the number of periods in a full seasonal cycle.

- For monthly data with annual seasonality: $m = 12$.
- For quarterly data with annual seasonality: $m = 4$.

Holt Winters Multiplicative

Component	Formula
Level	$L_t = \alpha \left(\frac{Y_t}{S_{t-m}} \right) + (1 - \alpha)(L_{t-1} + b_{t-1})$
Trend	$b_t = \beta(L_t - L_{t-1}) + (1 - \beta)b_{t-1}$
Seasonality	$S_t = \gamma \left(\frac{Y_t}{L_{t-1} + b_{t-1}} \right) + (1 - \gamma)S_{t-m}$
Forecast	$F_{t+h} = (L_t + h \cdot b_t) \cdot S_{t+h-m(k)}$

Forecasting Metrics

Mean absolute error

$$MAE = \frac{1}{n} \sum_{i=1}^n |A_i - F_i|$$

- **Advantages:**

- **Easy to Interpret:** The result is in the same units as the original data, making it very intuitive to understand (e.g., "the average error is 10 units").
- **Robust to Outliers:** Unlike metrics that square errors, MAE is not heavily influenced by a few large, isolated errors.

Day	Actual Temperature (°C)	Predicted Temperature (°C)	AE	Squared error
1	20	21	1	1
2	22	20	2	4
3	25	35	10	100
4	23	24	1	1
5	21	22	1	1
		MAE	3	21.4

MSE- Mean Squared Error

$$MSE = \frac{1}{n} \sum_{i=1}^n (A_i - F_i)^2$$

•Advantages:

- Penalizes Large Errors:** This is its main strength. It is a good metric to use when large errors are particularly undesirable or dangerous to the business.
- Mathematically Convenient:** The squaring function is differentiable, which makes MSE a popular loss function for optimizing models using gradient-based methods.

•Disadvantages:

- Not Interpretable:** The result is in squared units of the original data, which can be difficult to interpret intuitively.
- Sensitive to Outliers:** Because it heavily penalizes large errors, a single outlier can dramatically inflate the MSE, making it a poor representation of overall model performance.

RMSE

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (A_i - F_i)^2}$$

- **Advantages:**

- **In Same Units as Data:** The result is easy to understand and compare against the actual values
- **Penalizes Large Errors:** Like MSE, it is useful when large errors need to be avoided.

- **Disadvantages:**

- **Sensitive to Outliers:** It inherits the sensitivity to outliers from MSE.
- **Scale-Dependent:** Both RMSE and MAE are scale-dependent, meaning they cannot be used to directly compare the performance of forecasts on datasets with different scales (e.g., comparing a forecast for a company's revenue in millions of dollars with a forecast for its monthly website visits in thousands)

MAPE

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{A_i - F_i}{A_i} \right|$$

Advantages:

- **Scale-Independent:** This is its most significant advantage. It allows for the comparison of forecast accuracy across different time series or datasets with different scales.
- **Easy to Interpret:** The result is a percentage, which is often very easy for stakeholders to understand (e.g., "the model is off by an average of 5%").

Disadvantages:

- **Undefined for Zero Values:** The formula is undefined when the actual value is zero
- **Skewed by Small Values:** When the actual value is very close to zero, even a small absolute error will result in a huge percentage error, skewing the overall metric.

Day	Actual Temperature (C)	Predicted Temperature (C)	MAE	MSE	MAPE
1	20	21	1	1	0.05
2	22	20	2	4	0.09
3	25	35	10	100	0.40
4	23	24	1	1	0.04
5	21	22	1	1	0.05
			3	21.4	12.64
			RMSE	4.6260134	

Q&A